Experiment No 4

Information Gathering Using TheHarvester

Aim

To demonstrate information gathering using TheHarvester in Kali Linux by collecting publicly available domain-related information for the specified target domain.

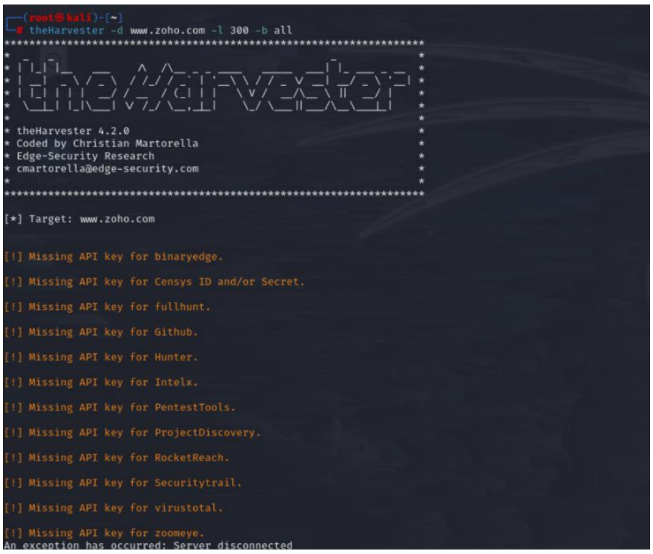
Algorithm

1. Start Kali Linux and open the Terminal.
2. Run TheHarvester with the selected target domain and search options.
3. Observe the information sources queried by the tool and the messages displayed in the terminal.
4. Review the final output for any results returned by the selected sources.
5. Record any connection, DNS, certificate, or missing-API-key errors shown during execution.

Output

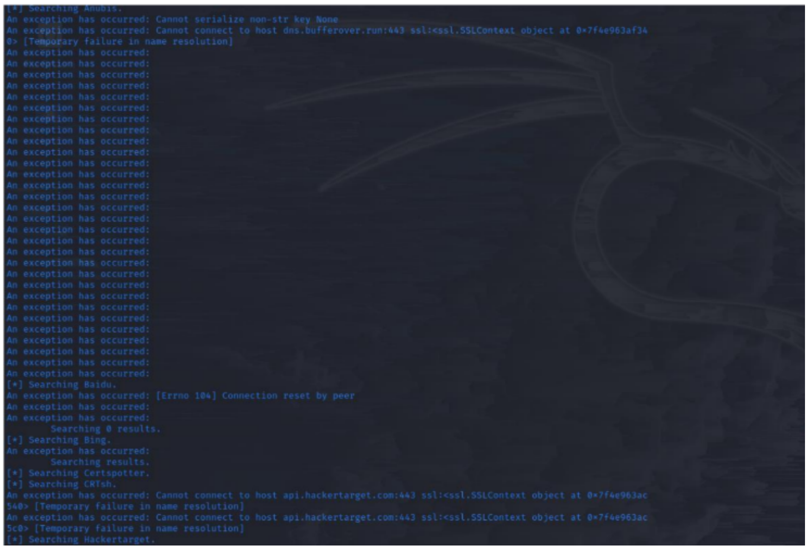
The following screenshots show the actual terminal output supplied for this experiment.

Output Screenshot 1



```
┌──(root㉿kali)-[~]
└─# theHarvester -d www.zoho.com -l 300 -b all
*******************************************************************
*  _   _                                            _             *
* | |_| |__   ___    /\  /\__ _ _ ____   _____  ___| |_ ___ _ __  *
* | __|  _ \ / _ \  / /_/ / _` | '__\ \ / / _ \/ __| __/ _ \ '__| *
* | |_| | | |  __/ / __  / (_| | |   \ V /  __/\__ \ ||  __/ |    *
*  \__|_| |_|\___| \/ /_/ \__,_|_|    \_/ \___||___/\__\___|_|    *
*                                                                 *
* theHarvester 4.2.0                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************

[*] Target: www.zoho.com


[!] Missing API key for binaryedge.

[!] Missing API key for Censys ID and/or Secret.

[!] Missing API key for fullhunt.

[!] Missing API key for Github.

[!] Missing API key for Hunter.

[!] Missing API key for Intelx.

[!] Missing API key for PentestTools.

[!] Missing API key for ProjectDiscovery.

[!] Missing API key for RocketReach.

[!] Missing API key for Securitytrail.

[!] Missing API key for virustotal.

[!] Missing API key for zoomeye.
An exception has occurred: Server disconnected
```

Output Screenshot 2



```
[*] Searching Anubis.
An exception has occurred: Cannot serialize non-str key None
An exception has occurred: Cannot connect to host dns.bufferover.run:443 ssl:<ssl.SSLContext object at 0x7f4e963af34
0> [Temporary failure in name resolution]
An exception has occurred:
An exception has occurred:
An exception has occurred:
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An exception has occurred:
An exception has occurred:
An exception has occurred:
An exception has occurred:
[*] Searching Baidu.
An exception has occurred: [Errno 104] Connection reset by peer
An exception has occurred:
An exception has occurred:
        Searching 0 results.
[*] Searching Bing.
An exception has occurred:
        Searching results.
[*] Searching Certspotter.
[*] Searching CRTsh.
An exception has occurred: Cannot connect to host api.hackertarget.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ac
540> [Temporary failure in name resolution]
An exception has occurred: Cannot connect to host api.hackertarget.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ac
5c0> [Temporary failure in name resolution]
[*] Searching Hackertarget.
```
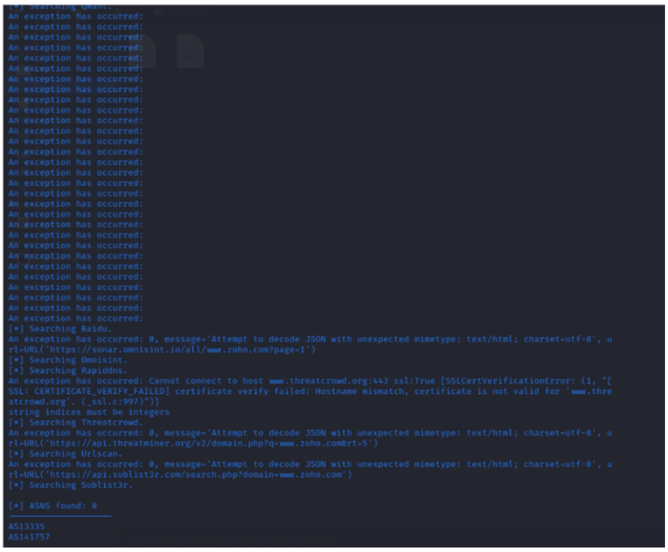

Output Screenshot 3



```
[*] Searching Qwant.
An exception has occurred:
An exception has occurred:
An exception has occurred:
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An exception has occurred:
An exception has occurred:
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An exception has occurred:
An exception has occurred:
[*] Searching Baidu.
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', u
rl=URL('https://sonar.omnisint.io/all/www.zoho.com?page=1')
[*] Searching Omnisint.
[*] Searching Rapiddns.
An exception has occurred: Cannot connect to host www.threatcrowd.org:443 ssl:True [SSLCertVerificationError: (1, "[
SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: Hostname mismatch, certificate is not valid for 'www.thre
atcrowd.org'. (_ssl.c:997)")]
string indices must be integers
[*] Searching Threatcrowd.
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', u
rl=URL('https://api.threatminer.org/v2/domain.php?q=www.zoho.com&rt=5')
[*] Searching Urlscan.
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', u
rl=URL('https://api.sublist3r.com/search.php?domain=www.zoho.com')
[*] Searching Sublist3r.

[*] ASNS found: 8
--------------------
AS13335
AS141757
```

Result

Thus, TheHarvester was executed in Kali Linux to demonstrate domain information gathering. The screenshots show the target domain and multiple lookup attempts; however, the run also reports missing API keys and network, DNS, or certificate-related errors, so the output does not confirm that all sources were queried successfully.